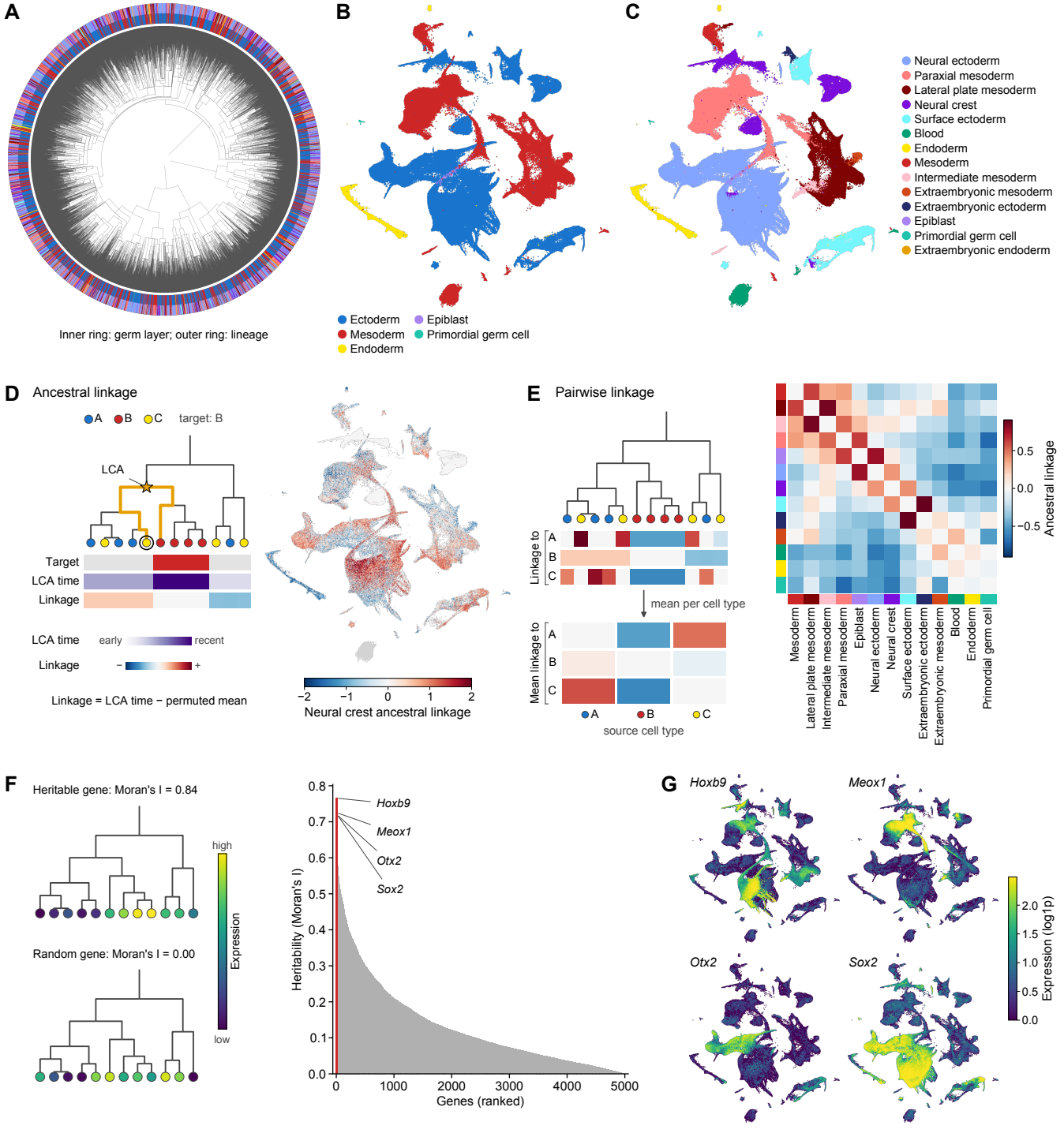

A
Inner ring: germ layer; outer ring: lineage
B
Ectoderm
Epiblast
Mesoderm
Primordial germ cell
Endoderm
C
Neural ectoderm
Paraxial mesoderm
Lateral plate mesoderm
Neural crest
Surface ectoderm
Blood
Endoderm
Mesoderm
Intermediate mesoderm
Extraembryonic mesoderm
Extraembryonic ectoderm
Epiblast
Primordial germ cell
Extraembryonic endoderm
D
Ancestral linkage
A B C target: B
LCA
Target
LCA time
Linkage
LCA time early recent
Linkage − +
Linkage = LCA time − permuted mean
−2 −1 0 1 2
Neural crest ancestral linkage
E
Pairwise linkage
Linkage to
A
B
C
mean per cell type
Mean linkage to
source cell type
Ancestral linkage
0.5
0.0
−0.5
Mesoderm
Lateral plate mesoderm
Intermediate mesoderm
Paraxial mesoderm
Epiblast
Neural ectoderm
Neural crest
Surface ectoderm
Extraembryonic ectoderm
Extraembryonic mesoderm
Blood
Endoderm
Primordial germ cell
F
Heritable gene: Moran's I = 0.84
high
Expression
Random gene: Moran's I = 0.00
low
Heritability (Moran's I)
0.0 0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8
Hoxb9
Meox1
Otx2
Sox2
0 1000 2000 3000 4000 5000
Genes (ranked)
G
Hoxb9
Meox1
Otx2
Sox2
Expression (log1p)
0.0 0.5 1.0 1.5 2.0